Distributed Commitment Escrow

A commitment nobody can see. That cannot be taken back. And that fires all at once, or not at all.

The parts

- **Locking a commitment:** Each company locks its commitment in an opaque envelope: everyone can see it exists, nobody can see what is inside, and the content cannot be swapped afterwards.
- **Coordinated reveal:** The envelopes open only when the threshold is reached, all together, and only after every reveal has been checked.
- **Public log against double commitment:** Everyone sees exactly the same list of commitments. A company cannot show one thing to one side and another to the other.
- **Time window and automatic return:** Whoever locked and vanished does not freeze everyone. When time runs out, everything is cancelled and returned.
- **Protection against misuse:** The mechanism is technically limited to one purpose: pausing a safety risk. Not price fixing, not anything else.
- **Scaling to N parties:** Not just two companies. Any number, with a flexible threshold, and no single party that everyone must trust.
- **Proving participation without revealing:** Anyone outside can verify that at least K companies committed, without knowing which, and without knowing to what.

Six decisions

- **A simple hash or Pedersen?** We chose Pedersen.
- **A trusted coordinator or one everyone audits?** We chose a coordinator everyone audits.
- **Free text or a closed purpose list?** We chose a closed list.
- **No time limit or a 72-hour window?** We chose a fixed time window.
- **Reveal one by one or all together?** We chose a simultaneous reveal.
- **An open list or a zero-knowledge proof?** We chose a zero-knowledge proof.